LAB 10/12 NOTES

$-\Delta_S u = f$ ON S

(WF) $\underbrace{\int_S \nabla_S u \cdot \nabla_S v \, dS}_{\text{STIFFNESS MATRIX}} = \underbrace{\int_S f v \, dS}_{\text{VECTOR OF } \partial_S ?}$

$K_{ij} = \int_S \nabla_S \phi_i \cdot \nabla_S \phi_j \, dS$

ON EACH PHYSICAL TRIANGLE $K = (A, B, C) \subset \mathbb{R}^3 \xleftarrow{\text{AFFINE MAP } \Phi} \hat{K} = (\hat{A}, \hat{B}, \hat{C}) \subset \mathbb{R}^3$

PHYSICAL TRIANGLE ON SPHERE (GLOB)

REFERENCE TRIANGLE (LOC)

$\downarrow$

BRING EVERYTHING IN A REFERENCE SPACE IS EASIER

WHY [$\overbrace{x}^{\text{GLOB}} = \Phi(\overbrace{\xi}^{\text{LOC}}) = A + \xi_1 \underbrace{AB}_{(B-A)} + \xi_2 \underbrace{AC}_{(C-A)}$ $\qquad \xi = \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}$

$\hat{\phi}_i = \phi_i \circ \Phi \qquad \phi_i = \hat{\phi}_i \circ \Phi^{-1}$

$\int_K g(x) \, dS = \int_{\hat{K}} g(\Phi(\xi)) \underbrace{|\mathrm{Jac}(\Phi)|}_{\parallel} d\xi$

2D IN 3D: $\sqrt{\mathrm{DET}(J^T J)}$

$J = [AB \quad AC] \qquad \sqrt{\mathrm{DET}(J^T J)} = \|AB \times AC\| = \dfrac{\mathrm{AREA}(K)}{\mathrm{AREA}(\hat{K})} = \dfrac{\frac{1}{2}\|AB \times AC\|}{\frac{1}{2}}$

$\mathrm{AREA}(K) = \int_{\hat{K}} |\mathrm{Jac}| \, d\xi = |\mathrm{Jac}| \; \mathrm{AREA}(\hat{K})$

$K_{ij} = \int_K \nabla \phi_i \nabla \phi_j \, dS \overset{\text{CHANGE VARS}}{=\!=} \int_{\hat{K}} (\nabla \phi_i(\Phi(\xi)) \cdot \nabla \phi_j(\Phi(\xi))) \, |\mathrm{Jac}| \, d\xi \overset{*}{=}$

$\nabla_\xi \hat{\phi}(\xi) = J^T \nabla \phi(x) \quad , \quad x = \Phi(\xi)$

$\hat{\phi}_i = \phi_i \circ \Phi$

$\nabla \phi_i = J (J^T J)^{-1} \nabla_\xi \hat{\phi}_i$

$\overset{*}{=} \int_{\hat{K}} (\nabla \phi_i(\Phi(\xi)) \cdot \nabla \phi_j(\Phi(\xi))) \sqrt{\mathrm{DET}(J^T J)} \, d\xi$

$K_{ij}^K = (\nabla \phi_i \cdot \nabla \phi_j) \int_{\hat{K}} \underbrace{\sqrt{\mathrm{DET}(J^T J)}}_{\|AB \times AC\|} d\xi = (\nabla \phi_i \cdot \nabla \phi_j) \underbrace{\tfrac{1}{2}\|AB \times AC\|}_{\mathrm{AREA}(K)}$